Embodied Provenance for Immersive Sensemaking

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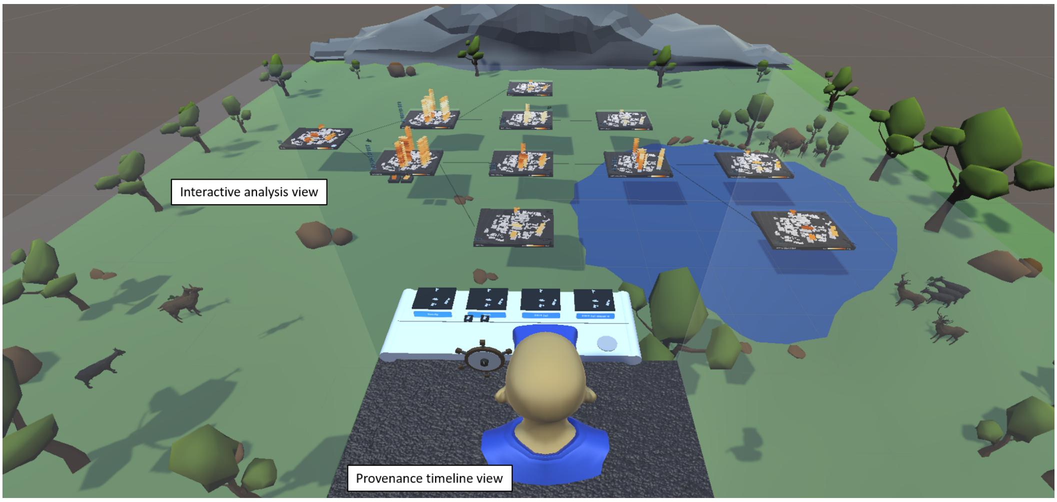

Fig. 1. We explore the concept of embodied provenance with a prototype implementation that allows users to interact with embodied data representations in a large spatial environment. Users can conduct complex analyses of spatio-temporal data in the interactive analysis view. In this figure we see a user in the control booth above, observing a previous analysis using the provenance timeline view.

Immersive analytics research has explored how embodied data representations and interactions can be used to engage users in sensemaking. Prior research has broadly overlooked the potential of immersive space for supporting analytic provenance, the understanding of sensemaking processes through users' interaction histories. We propose the concept of embodied provenance, the use of three-dimensional space and embodied interactions in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We design a conceptual framework for embodied provenance by highlighting a set of design criteria for analytic provenance drawn from prior work and identifying essential properties for embodied provenance. We develop a prototype system in virtual reality to demonstrate the concept and support the

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conceptual framework by providing multiple data views and embodied interaction metaphors in a large virtual space. We present a use case scenario of energy consumption analysis and evaluated the system through a qualitative evaluation with 17 participants, which show the system's potential for assisting analytic provenance using embodiment. Our exploration of embodied provenance through this prototype provides lessons learnt to guide the design of immersive analytic tools for embodied provenance.

CCS Concepts: • **Human-centered computing** → **Virtual reality**; **Interaction techniques**; **Visualization**.

Additional Key Words and Phrases: embodied provenance, immersive analytics, sensemaking, spatial memory

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1 INTRODUCTION

Immersive analytics has brought rich opportunities for the exploration of data visualisations through engaging embodied interactions. Embodiment has been used in recent research to support sensemaking, by representing abstract data as virtual objects to enable easy manipulation and intuitive interactions. For example, ImAxes [10] allows users to construct ad-hoc visualisations by combining individual axes, while TimeTables [50] allows users to 'drill down' into temporal data by reaching into the data structure to pull out layers and stretch their time dimension. Recent works have also taken advantage of large spatial environments afforded by Virtual Reality (VR) for organising and browsing embodied data constructs (e.g., DataHop [20], TimeTables [50], GestureExplorer [27]), to further support sensemaking.

Sensemaking is the process of extracting information and building understanding from a collection and organisation of data [1, 28]. More recently, researchers have come to realise that collecting and reflecting on *analytic provenance* information — knowledge about the history of users' interactions and reasoning through this complex reasoning process [48] — is in itself an important part of sensemaking [44, 47]. However, to date, there has been little exploration of supporting provenance in immersive analytics applications.

Building on the achievements of immersive analytics, we believe that embodiment and immersive space can be used for supporting analytic provenance. Previous research in immersive analytics found that embodiment intervention and spatial distributions reduce cognitive load and shows promises to aid spatial memory [4, 26]. Large volumes of provenance data may be generated when conducting complex analysis tasks, and the use of embodiment could help to reduce users' cognitive load [25, 26] by externalising the data into the spatial environment. To our knowledge, there has not been any systematic attempt for immersive analytics provenance using spatial organisation of embodied data constructs and embodied interaction metaphors.

Building on our prior work [51], we propose the concept of *embodied provenance*, which we define as the use of three-dimensional space and embodied interactions in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We aim to build on analytics provenance and immersive analytics work by establishing a conceptual framework for embodied provenance. We highlighted a full set of criteria drawn from prior work on analytic provenance and identified three key properties for embodied provenance. We further implemented a prototype system and presented a use case scenario to demonstrate the concept in VR, by providing multiple data views, spatial environments, and novel embodied interactions. Qualitative evaluation of the prototype system with 17 participants shows the potential and promise of our concept. This research extends our understanding of how to leverage embodied data constructs and embodied interactions to facilitate the analysis of sensemaking processes' histories.

2 RELATED WORK

2.1 Embodiment and Spatial Memory in Immersive Analytics

The term *embodiment* is broadly defined as a tangible or visible representation of an abstract idea. In the realm of VR, the term often refers to the synchronisation of a virtual body with the physical body of a user [15]. Dourish [12] furthers the definition of embodiment to describe the natural and significant presence of digital artefacts in a way they connect to the situated environment and provide intuitive interactions. Researchers in Immersive Analytics [7, 32] have applied this definition to encompass embodied data artefacts or controls for manipulating data. For instance, ImAxes [10] supports direct operations to assemble and disassemble the axes of 3D coordinates to visualise multivariate data. The immersive space-time cube [45] offers embodied interactions to manipulate 3D spatio-temporal visualisations. Tilt Map [49] enables embodied transitions between 2D and 3D map visualisations. Embodied hand gestures also have been used to explore 3D graph visualisations [22] and augmented reality (AR) maps [34]. Their findings suggest that embodied visualisations and interactions benefit data understanding and can reduce cognitive load.

Externalising thoughts into the environment is another way to offload cognition [8, 36], also has shown promises to aid spatial memory [2, 3, 9, 38]. Previous research in immersive analytics found that embodiment intervention reduces cognitive load and increases the capacity of spatial memory [26]. Batch et al. [4] found that embodied interactions and spatial distributions reduce cognitive load in immersive data analysis. Liu et al. [29] found that different layouts and organisations of 3D visualisations require different physical navigation, leading to diverse support for spatial memory. SuperSight [39] synthesises auditory, visual and positional information to boost users' memory in immersive environment. While many of these examples demonstrate sensemaking processes in immersive analytics, the analytic provenance aspect remains under-explored. Next section describes the importance of analytic provenance and a summary of existing works in this domain.

2.2 Analytic Provenance for Sensemaking

Sensemaking is the process by which people build understanding and extrapolation from a collection and organisation of data [1, 28]. When undertaking complex sensemaking tasks, it is difficult to maintain the mental model of the process after a long period of exploration and analysis [35]. Therefore, it is valuable to maintain the history of user interactions and reasoning process, known as analytic provenance [48]. Analytic provenance can be used for various purposes [37], such as recalling the analysis process [46], reproducing analysis workflow [13, 43] and collaboration. For example, GRASPARC supports easy rollback or branch the analysis history using a history tree [6]. SenseMap and Trrack [11, 35] support collaborative sensemaking and communication by sharing the analysis process. InsideInsights [33] makes full use of annotations to allows for iterative structuring of insights into hierarchies linked to analytic provenance in a collaborative setting. An interaction model with annotations is also presented to support collaborative discovery and recall of past explorations [19]. Sarvghad and Tory used multiple views for supporting asynchronous collaborative exploratory visual data analysis from a dimension coverage perspective on visualization history [40]. HARVEST [17] presents a trail with the logical history of user performed actions for supporting insight provenance in collaborative analysis. 2D space is often used in supporting analytic provenance, such as the InsideInsights system [33] mentioned earlier. Furthermore, systems such as ExPlates [24] can spatialise analysis steps on 2D canvas in a hierarchy structure. Studies in analytic provenance show that keeping track of the analysis history and representations of this history provide benefits for complex sensemaking tasks[44, 47].

However, most research on analytic provenance is browser-based. To our knowledge, little study has systematically investigated how analytic provenance can be represented and used for supporting sensemaking tasks in immersive environments. In this paper, we focus on immersive analytic provenance with embodiment to support sensemaking.

3 CONCEPTUAL FRAMEWORK OF EMBODIED PROVENANCE

Our work is, to our knowledge, the first to consolidate immersive analytics with analytic provenance domains by defining a conceptual framework that we call embodied provenance. Our proposed concept is a combination of five key design criteria for analytic provenance and three embodied provenance properties, as outlined and illustrated below.

3.1 Design Criteria for Analytic Provenance

We first reviewed existing concepts in traditional analytic provenance for 2D visual analytics. Ragan et al. organised purposes of provenance under recall, replication, action recovery, collaborative communication, presentation, and meta-analysis [37], and Cutler et al. presented several design goals based on these purposes, such as supporting action recovery, supporting reproducibility, supporting annotation [11]. Heer et al. concludes operations on history as navigation to states in the history, editable histories, metadata and annotation, search and filter, and export for communication [21]. Based on this prior work, we highlight the key design criteria for analytic provenance:

AP1. Support Memory Recovery: The system should support users to recall their previous analysis. This recovery of process memory should enable awareness and understanding of what analytic tasks have previously been completed, what findings have been established, and what tasks remain to be completed in the future [21, 37].

AP2. Support Action Recovery: The system should support users to maintain the action history that allows undo / redo operations during the analysis process. Users should have the freedom to conduct exploratory analysis and the ability to correct the human errors and return to previous state of analysis through action recovery [11, 21, 37].

AP3. Support Replication and Reproducibility: The system should support users to replicate and reproduce the steps or workflow of a previous analysis. Replication pertains to the analysis, whereas reproducibility is primarily concerned with the ability to obtain consistent results of the analysis. For instance, users should be able to conduct branching investigations or revise the analysis without modifying the previous states [11, 37].

AP4. Support Annotation: The system should allow for a means of note taking during the analytical process and for post-analysis activities. This could include techniques such as highlighting visualisation, adding text comment, or voice memo. These annotations should enable users to keep track of significant insights in the context in which they were found and to communicate their analysis to others.[11, 21, 37].

AP5. Support Sharing and Collaboration: The system should enable users to communicate and share insights in synchronous or asynchronous collaborative analysis settings. This collaborative analysis should allow for information sharing such as provenance history or annotations [11, 21, 37]. Furthermore, the sense of presence of others is also an important factor in embodied provenance, which can be achieved by showing avatars of other users.

3.2 Essential Properties for Embodied Provenance

While the theoretical framework of analytic provenance in general (as described above) has been developed, the differentiating factors of analytic provenance in immersive sensemaking lack of theoretical foundation. We draw inspiration from previous work on spatial interaction and embodied interaction. For instance, DataHop [20] takes advantage of a large environment afforded by VR to encode interaction history as three-dimensional, room-sized graph structure. Conversely, ImAxes [10] uses direct manipulation of embodied graph axis for intuitive interaction with high-dimensional data. TimeTables [50] combines both approaches, using direct manipulation of space-time cube elements to create persistent views that capture the analysis history. In our quest toward a holistic conceptual framework of embodied provenance, we build on existing immersive analytics work and propose the following properties:

EP1. Use embodied representation of data construct and interaction metaphor. Embodied representation of both data constructs and interaction metaphors is crucial in embodied provenance. Instead of visualising the analysis history as abstract representations, embodied provenance approach would represent events as embodied three-dimensional representations. For example, as virtual tabletop charts [50], or building-like network of 3D charts [20]. Embodied provenance also includes the use of visual metaphors of physical objects to represent the system's component and interactions with embodiment features. For instance, using 3D axes for arranging attributes in a virtual shelf [10, 29]. This property fosters intuitive and fluid interaction[14] – persistent elements are created implicitly through embodied manipulation of visualisation constructs – with elements in immersive sensemaking systems. For example, use tabletops as nodes and integrate the selection and manipulation of the data layer with the creation of a 'zoomed-in' view of the layer in the TimeTables scenario [50].

EP2. Support seamless integration between temporal and three-dimensional spatial aspects. Temporal aspect (e.g., history of events) and spatial aspects (e.g., representations of this history) are important factors in analytic provenance. Traditionally, the amount of temporal information presented is subject to the display space budget, meaning the layout and type of event history representation is constrained to a limited 2D space. This is not the case for embodied provenance environments with vast display spaces. Embodied provenance makes use of this space to visualise great amount of historical information. For example, lay out historical analysis steps in the large virtual environment [20, 27, 50]. Integrating temporal aspect to three-dimensional space might also support information acquisition as people generally have a better sense of their surroundings, i.e. environment awareness[23].

EP3. Support environmental features to aid spatial memory. As described earlier, analytic provenance processes should support memorability. Through support for strong sense of presence, embodied provenance offers a means to support this criteria that is beyond traditional analytic provenance on 2D displays. While exploring a 3D world, people gather information of their surroundings and build clues that are embedded in environmental features. Furthermore, spatial locations are integrated with events that occurred there in the process known as episodic memory formation [41] (for instance remembering where you were when a significant event occurred). Thoughtful design of recognisable environmental features can potentially support both spatial awareness triggering of episodic memories that previously occurred in certain locations, aiding memorisation of analytic steps by recalling locations.

3.3 Illustrating the Framework

In order to demonstrate the conceptual framework, we use the five analytic provenance design criteria (**AP1–5**) and the three key properties for embodied provenance (**EP1–3**) to compare and analyse eight existing immersive and provenance systems that motivated this work (see Figure 2). We made this comparison based on the published papers, and acknowledge that the systems may have new features or ones we are not aware of.

Most immersive analytics systems in the table, i.e. DataHop [20], TimeTables [50] and Gesture-Explorer [27], make use of embodied data representations and interaction metaphors (EP1), and

	System	Design Criteria for Analytic Provenance (AP)					Properties for Embodied Provenance (EP)		
		AP1	AP2	AP3	AP4	AP5	EP1	EP2	EP3
Immersive Analytics	DataHop [20]	Analysis steps maintained, but can't recall tasks completed	Can return to previous state, but can't undo/redo	Analysis can be reused by copying and pasting Filters			Data is represented as network of 3D charts	Large 3D space is used for analysis histories	
Immersive Analytics	TimeTables [50]	A storage wall helping to recall analysis histories	Can return to previous state, but can't undo/redo				Data is represented as virtual tabletops	Large 3D space is used for analysis histories	A room and walls used, but no highlighted landmarks
Immersive Analytics	GestureExplorer [27]	Analysis steps maintained, but can't recall tasks completed	Can return to previous state, but can't undo/redo		Gesture marking for annotation		Data is represented as network of gestures	Large 3D space is used for analysis histories	
Immersive Analytics	ImAxes [10]						Physical 3D Axes are used for interaction		
Provenance	Trrack [11]	Analysis histories maintained showing tasks completed	Can undo/redo the current node	Graphs can be stored for reproducing	A user-defined string for annotation	Allow sharing by sending URL			
Provenance	SenseMap [35]	Analysis histories maintained showing tasks completed	Can return to previous state, but can't undo/redo	Multiple copies of knowledge maps can be created	A written note can be created	Maps can be saved and loaded for sharing			
Provenance	ExPlates [24]	Analysis histories maintained showing tasks completed			A comment field and separate annotation plate				
Provenance	InsideInsights [33]	Analysis histories maintained showing tasks completed	Can return to previous state, but can't undo/redo		Annotation cells can be created	Allow sharing a link to the document			
	Our prototype	Analysis histories could be replayed	Can undo/redo last action	Analysis can be reused by copying and pasting tabletops	Speech-to-text annotations	Analysis can be replayed for sharing	virtual tabletops and physical tools are used	Large 3D space is used for analysis histories	Six themed environments with sounds and landmarks

Fig. 2. Overview of the design criteria and the key properties for embodied provenance as emphasised in a sample of immersive analytics and provenance systems. Immersive analytics systems are shown in pink cells and provenance systems are shown in yellow cells. Design criteria for analytic provenance are shown in the blue columns, and properties for embodied provenance are shown in the green columns. The brightness of the color indicates our level of confidence (darker being more confident).

integrate analysis history into a large 3D virtual space (EP2). While we observed a limited use of environmental features to assist spatial memory for analysis (EP3), these systems support memory recovery (AP1) and action recovery (AP2) to some extent. ImAxes [10], one of the systems that inspired us with embodiment, doesn't clearly support any design criteria of analytic provenance and only has the property for embodied provenance of 3D axes metaphor (EP1). Although the system takes full advantage of 3D space, the temporal information it presents is not analysis history, therefore, the property of EP2 which requires both analysis history and the use of 3D space, is not satisfied. Similarly, even though some traditional 2D provenance systems, such as ExPlates [24] and InsideInsights [33], spatially lay out the temporal histories on 2D displays, they don't have the property of EP2 (3D spatial aspects). The provenance systems in the table meet the design criteria of analytic provenance with a weak emphasis on AP3 supporting replication and reproducibility, however, they don't have any properties for embodied provenance due to the traditional 2D nature.

In our paper, we present a prototype system that aims to meet all design criteria and properties for embodied provenance. We will refer back to the design criteria (**AP1–5**) and the key properties for embodied provenance (**EP1–3**) when describing the system visualisations and interactions in the next section.

4 PROTOTYPE

Learning from the few examples of visualisation systems reviewed, we aim to explore how such an embodied conceptual framework can be instantiated as a prototype system that satisfies all criteria. We decided to extend the TimeTables system because of the similarity in use cases (building energy data, described in Section 5) with ours and the availability of the code provided by the authors. We keep the core functionalities from TimeTables in our prototype system for data exploration except storing tables as 2D maps on the wall and jumping in and out of the table. The interactions we kept include filtering a time layer or block by grabbing the object to make a copy, and stretching a layer or block to increase the level of time granularity, for example, from one year to monthly data. On this basis, we looked deeper into the features that better support embodied provenance. In the next sections, we describe the implementation, visualisation design, and interaction design of our prototype system.

4.1 Implementation

The system is built in Unity3D[1] (version 2020.3.17f1) and displayed with an Oculus Quest 2 VR device. A Windows 10 PC equipped with an Intel i7 processor and an NVIDIA GeForce GTX 1080 graphics card is used to run the prototype. We use Mixed Reality Toolkit[2] (MRTK) for implementing the interactive components and hand tracking. Mapbox Unity SDK[3] is installed for the campus map on the tabletop.

4.2 Visualisation

We provide various embodied data visualisations to support analytic provenance through multiple data views and large virtual environments with different themes in our system.

4.2.1 Multiple Views. The system provides multiple views to support embodied provenance inspired by Shrinivasan and van Wijk's framework [42], SenseMap [35] and the CLUE model [18]. To support analytical reasoning process, Shrinivasan and van Wijk proposed a data view for the actual visualisation of the data, a navigation view showing the exploration process in a tree, and a knowledge view to record findings with notes and link them to the visualisation states. Similarly, SenseMap includes a browser view, a history map and a knowledge map, while in the CLUE model, a provenance view and a story view are generated upon the exploration of the data visualisation. We combine and refine these concepts based on our motivation and design goals to create the following three views:

- **Interactive analysis view**: actual analysis with visualisations in a hierarchy tree structure
- **Mini map view**: an overview of the current visualisation state
- **Provenance timeline view**: analysis histories in a linear timeline structure

The interactive analysis view consists of all the visual representations and interactions for the data exploration and analysis with the surrounding environment. The visualisations are similar to those in TimeTables [50] which consists of colour encoded layers and blocks representing temporal data above virtual tabletops and pedestals respectively **(EP1. Use embodied representation of data construct and interaction metaphor)**. However, unlike the TimeTables system where users can place the tabletops at any preferred position in the space, the layout of the tabletops in our system is a hierarchy tree as tree-like visualisations are usually used to depict the branching trace structure [16] of an analysis process. The nodes are the tabletops in different levels of time granularity (year, month, week, day and hour) and links connect between a tabletop and its child tabletop in the next detailed level of time granularity **(EP2. Support seamless integration between temporal and 3D spatial aspects)**. Using this layout, users can avoid the visual clutter and easily trace the relationship of different tabletops and the provenance data.

The mini map view includes a scaled and simplified interactive analysis view without the interactions for analysis. The mini map view changes in correspondence with the interactive

[1] https://unity.com/
[2] https://developer.leapmotion.com/mrtk-unity
[3] https://www.mapbox.com/

analysis view to reflect recent changes and provides an overview of the current visualisation state. A miniature avatar indicates the user's position in the mini map view so that they can obtain a spatial awareness of where they are located in the environment. Users can also easily navigate to a chosen tabletop in the interactive analysis view by tapping on the thumbnail tabletop in the mini map view.

The provenance timeline view contains an embodied timeline to support analytic provenance tasks. A timeline representation is chosen because previous research shows that linear temporal sequences provide a clear sense of the direct history of the exploration [16]. The timeline consists of multiple thumbnail tabletops, every time a new tabletop is created in the interactive analysis view, a thumbnail tabletop will be appended to the timeline based on the time they were created **(EP2. Support seamless integration between temporal and 3D spatial aspects)**. Following the metaphor of a control booth in a theatre (which provides technicians with an overview of the stage), the timeline is located above the interactive analysis view where the analyst has access to a complete view of the visualisations and exploration process in the interactive analysis view below. To manage the length of the timeline, only four thumbnail tabletops can be shown at one time. The timeline combines the metaphors a conveyor belt and a ship's helm — users can rotate an embodied helm to move the timeline conveyor forward or backwards.

4.2.2 Themed Spatial Environments. In the physical world, we rely on unique features of our surrounding environments as landmarks for navigating and recalling locations. According to Lynch [30, 31], landmarks are more memorable if they have a clear form, contrast with their background, and are in a prominent spatial location. In contrast to the large empty environment in TimeTables, our system allows users to choose from six themed environments with spatial landmarks and audio aimed to facilitate memory recall **(AP1. Support Memory Recovery and EP3. Support environmental features to aid spatial memory)**. Each environment has its distinct scene and background sound. Within each environment, different landmarks are presented in the space, for example, Christmas trees, snowmen, a bridge decorated with coloured lights, benches and streetlights are shown at different locations in the Christmas scene.

4.2.3 Tool Menu. We support embodied interactions using a tool metaphor [5] **(EP1. Use embodied representation of data construct and interaction metaphor)**. A tool menu is shown when users face the palm toward them. The tool menu consists of three side-by-side buttons for a lever for undo and redo, a mini map and a microphone respectively from left to right. We replaced the previously used containers with buttons based on the feedback from the Video group in the evaluation (discussed in Section 6) so that users can simply press the buttons to show and hide the tools rather than grabbing and dropping. The tool menu also includes a button to allow quick navigation to the timeline.

4.3 Interactions

The system provides multiple physical interactions **(EP1. Use embodied representation of data construct and interaction metaphor)** in supporting embodied provenance.

4.3.1 Replay-It. The Replay-It feature allows users to replay their previous exploration processes, to help them recall what analyses have been done and the findings that were discovered **(AP1. Support Memory Recovery)**. The playback includes all visualisations created, annotations for insights, and an avatar representing the user's previous position, head movements and hands movements. In addition to analyses conducted by themselves, users are also able to watch the replay of analysis processes conducted by other analysts **(AP5. Support Sharing and Collaboration)**. From the timeline view, users can start the replay by selecting the tabletop on the timeline, the system will

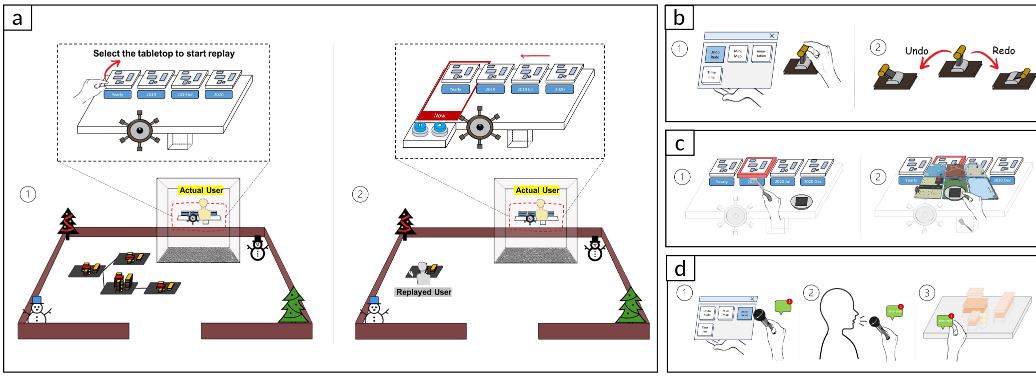

Fig. 3. Embodied interactions provided by the prototype. a) Replay-It, b) Do-It-Lever, c) Copy-It-Spatula, d) Note-Down-Mic.

then replay the analysis starting from the creation of the selected table in the interactive analysis view. Users can choose to stay in the control booth above the interactive analysis view to watch the replay from an overview perspective, or navigate down into the environment and look over the avatar's shoulder to get more detailed information by pressing a 'Go Down' button on the timeline. During the replay, the timeline will scroll along and an area highlighted with red light will indicate the current point in time. A pause button and a stop button are also provided on the timeline so that users can pause/play or stop the replay process. Figure 3a) shows how the Replay-It interaction works.

4.3.2 Do-It-Lever. We provide an embodied lever to allow undo and redo interactions which enable users to achieve the action recovery along with their analysis process **(AP2. Support Action Recovery)**. Users can pull a lever opened from the tool menu to the left to undo the last action while pull the lever to the right to redo the action (see Figure 3b)). The supported actions include creating a tabletop for multiple buildings exploration, creating a pedestal with a block for the analysis of a single building, stretching a layer above the tabletop and stretching a block on the pedestal for a more detailed time granularity.

4.3.3 Copy-It-Spatula. The Copy-It-Spatula enables users to copy created tabletops and pedestals and paste them in another environment (see Figure 3c)). This feature allows users to reproduce the steps of a previous analysis process **(AP3. Support Replication and Reproducibility)**, or to branch off from analysis created by another user **(AP5. Support Sharing and Collaboration)**. When users grab the spatula from a container under the timeline, a plate for containing the copied objects will appear. Users can copy the tables or pedestals on the timeline to the plate by shoveling them with the spatula. Users can then paste the copied visualisations into a new environment by dropping the plate onto one of the presented six environments. Users will be automatically teleported to that environment alongside the copied tabletops and pedestals. A mini version of the environment where the tabletop is copied from is created on the right corner of the tabletop, and users can navigate back to the source scene by selecting the mini environment.

4.3.4 Note-Down-Mic. The Note-Down-Mic provides users the ability to add annotations throughout the analysis process **(AP4. Support Annotation)**. The annotations can be ideas, comments, findings or insights. Users can add them to the analysis conducted not only by themselves, but also by other analysts **(AP5. Support Sharing and Collaboration)**. Users can open a microphone

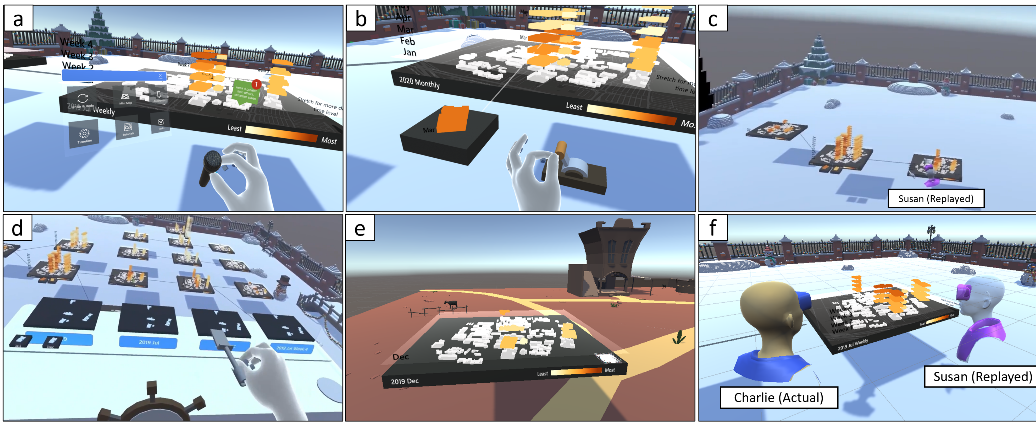

Fig. 4. Use case scenario. a) Grab the microphone to add annotation for insights, b) Pull the lever to left to undo from the blocks for weekly view to a block for one month, c) Replay of previous analysis conducted by Susan for her own investigation (from the timeline view), d) Copy a specific tabletop using the spatula and e) paste it into a new environment for further investigation, f) Replay of previous analysis conducted by Susan for Charlie's exploration (from the interactive analysis view)

from the tool menu and speak into it. Their words will be converted to text shown on a notification note and users can place the note near the object to be annotated (see Figure 3d)).

5 ENERGY DATA SCENARIO

We present a use case scenario to show how our system can support the framework of embodied provenance. In our scenario, Susan and Charlie, as data analysts, would like to analyse historical energy consumption data of five buildings from 2019-2020 on a university campus. The video for this scenario is included in the supplementary materials.

Susan first focuses on the exploration of data in 2019 (before the outbreak of Covid-19). She investigates the monthly data in 2019 and compares the data in July and December by creating relevant tabletops. She finds energy consumption in July is much higher than that in December and identifies a peak value in week 4 after a further investigation of the data in July. She speculates that this may because a new semester starts in week 4 of July. She then uses the Note-Down-Mic to annotate a note containing this finding and her inference to the table, for sharing with other analysts **(AP4. Support Annotation)** (see Figure 4a). She further explores the data in week 4 of July and compares the difference of energy consumption in weekdays and weekends by dragging out more tabletops.

Susan next moves to the exploration of data in 2020 and would like to look at the data in April for a building used for teaching purposes when a lockdown started in April 2020. However, she mistakenly extracts the data in March for that building and further investigates the data for a more detailed time level. After realising this mistake, she uses the Do-It-Lever to undo the actions **(AP2. Support Action Recovery)** (see Figure 4b).

After a long session of exploration and analysis, she has created a large hierarchy of 15 tabletops along with multiple pedestals for data of certain buildings. She would like refer back to an analysis step she did for July 2019. She can only remember that there is a snowman in the environment in front of the tabletop but forgets exactly which tabletop it was and the specific analysis she did. So she uses the Replay-It feature to view the playback of the previous analysis process and successfully finds that specific step **(AP1. Support Memory Recovery)** (see Figure 4c).

Now she would like to focus on the tabletop for December 2019 to see if she can find any interesting patterns. However, she doesn't want to modify the existing tabletops in the current environment, so she copies the tabletop using the Copy-It-Spatula and pastes it to a new environment for further investigation **(AP3. Support Replication and Reproducibility)** (see Figure 4d and e).

After a week, Susan goes back to the environments to review her analysis and she can easily understand her overall analysis strategy by seeing how the tables showing different time scales and locations are arranged and linked together. She can easily refresh her memory with the help of the annotations and the replay. She then conducts some further analysis and later invites her colleague Charlie to review the analysis and confirm her findings.

Charlie enters Susan's environments and quickly understands the previous analysis done by Susan via revisiting those steps using the replay feature (see Figure 4f) and investigating the annotations. He discovers an interesting pattern overlooked in the analysis by Susan, so he copies the relevant branch and pastes it to a new environment to further investigate whether the pattern continues across a later time frame **(AP5. Support Sharing and Collaboration)**.

6 QUALITATIVE EVALUATION

We conducted a qualitative evaluation to collect feedback on our prototype system. We are particularly interested in evaluating the usefulness of the proposed visualisations and interactions in meeting the design criteria for supporting embodied provenance. In our study implementation, annotations are recorded through a Wizard-of-Oz approach where an experimenter manually types in the dictated note.

6.1 Participants

We recruited a total of 17 participants in two groups:

- **Video group**: We invited five remote and local experts in relevant fields to evaluate the prototype system using video clips. Participants include one associate professor (P1) holding expertise in provenance and visualisation, two data analysts (P2-P3) working on energy data visualisation at a university, one research fellow (P4) and one final-year PhD candidate (P5) in the domains of immersive analytics, VR/AR and human-computer interaction.
- **In-person study group**: We recruited another 12 participants (P6 - P17) to evaluate the system based on their direct first-hand experience in VR environment (8 males and 4 females). Participants have multiple dicipline expertise in the areas of immersive analytics (5 participants), provenance (1), data visualisation (5), data analysis (4), VR/AR (8) and human-computer interaction (9).

6.2 Procedure and Analysis

The evaluation was conducted remotely via video conferencing for remote experts in **Video group**. The user study was conducted in person for all participants in **In-person study group** three months after the Video group. The evaluation lasted about 1 hour per participant and consisted of the following stages.

Introduction and Overview: Participants were introduced briefly to the background of our research, the embodied provenance framework, the visualisations and interactive features implemented in our prototype system. Participants were asked to fill in a pre-questionnaire to provide basic information such as age and gender, and their areas of expertise.

System Evaluation: For participants in **Video group**, we asked them to review the system by presenting short videos of a potential use case of campus energy consumption data analysis (5 minutes long) and all features of our system (longest being 2 minutes). Videos were used to ensure that both remote and local experts received the same content. Whilst the video was playing, we described the use case and the interactions verbally to the participants. They could also pause or replay the videos at any time or ask for clarification whenever necessary.

For participants in **In-person study group**, we instructed them on necessary hand gestures and provided them training videos in the virtual environment. The training includes basic interactions for data exploration, teleportation and features for embodied provenance (Replay-It, Do-It-Lever, Copy-It-Spatula and Note-Down-Mic). The training videos are provided in supplementary materials. Participants were asked to follow the videos and the experimenter described the interactions verbally if a participant had difficulty performing the actions. The training continued until they could interact with the system without help. After participants completed the training session, they entered an VR environment where some analyses were already conducted by the experimenter. Participants were asked to answer eight questions about the data and the analysis process. The first four questions need to be answered by watching a playback of the experimenter's analysis using Replay-It feature. The questions include finding which month has higher energy consumption and what the experimenter thought the reason is by reading the annotation, what analysis did the experimenter conduct after adding the annotation and which time period did the experimenter delve into first. After answering the first four questions, participants could start their own exploration and complete the last four questions on comparing energy consumption in different time periods, comparing energy consumption for different buildings, adding annotations for any findings, copying and pasting certain tables to a new environment for further exploration.

Questionnaire and Feedback: Participants were asked to complete a questionnaire after evaluating the system. The questionnaire consisted of 5-point Likert scale questions (1 – Not at all, 2 – Slightly, 3 – Somewhat, 4 – Moderately, 5 – Extremely) on the usefulness of each visualisation and interactive feature in supporting related design criteria, as well as the usability of the system in supporting embodied provenance. After each Likert scale question, a follow-up open-ended question was provided for brief explanation of their choice. We also included open-ended questions in the questionnaire to collect participants' general feedback on other potential use cases and improvements of the system. The participants in both groups answered the same 10 open-ended questions. We asked the participants in the in-person study group an additional question regarding their experience with the system.

Data Analysis: We conducted thematic analysis on the responses of the open-ended questions. We imported all responses into sticky notes on a digital Miro board. There are 182 cards in total. The lead researcher then read through these cards and generated initial codes. Then, we used these codes to define different themes. We created themes within each question. Most questions have different themes while we also observed small number of overlaps. There are 50 themes in total, with a range of 7-11 themes per question. Further details are provided in the supplementary material.

6.3 Results

Overall, the responses we collected from the feedback questionnaire were positive (as seen in Figure 5). Participants generally agreed that our prototype system is useful in supporting embodied provenance (5 of 17 participants rated it as extremely useful, 9/17 moderately useful, 3/17 somewhat useful).

The remainder of the results are the outcome of our thematic analysis. Participants agreed that the system can be generalised to any immersive visual analytics scenario, especially for spatial-temporal data. We summarise the related feedback on each criteria **(AP1-5)** and key properties for embodied provenance **(EP1-3)** below.

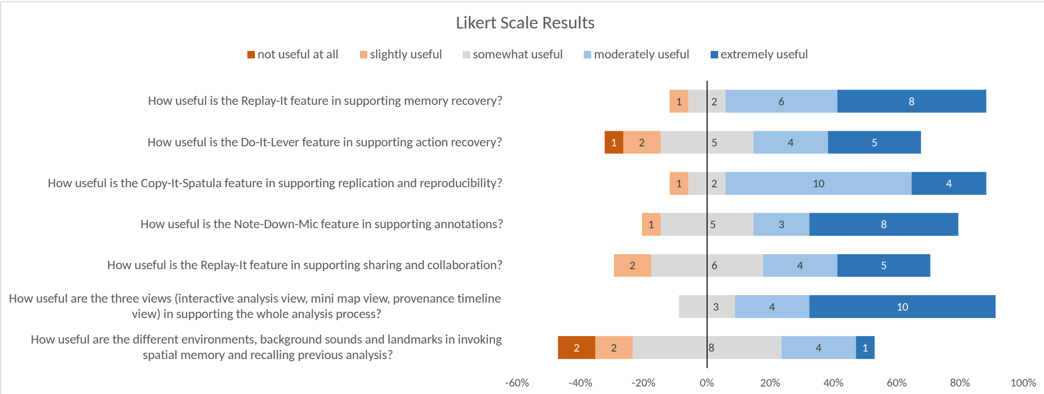

Fig. 5. The results of Likert scale questions

AP1. Support Memory Recovery: Most (12/17) participants reported that the Replay-It feature is helpful for recalling previous analysis. Seven participants mentioned the use of the playback. Among these participants, 5 of them who participated in the **In-person study group** confirmed that watching the playback is helpful. One of these 5 participants suggested more controls on the playback such as adding fast forward or slow down options. Two participants in the **Video group** hesitated about the use of playback/animation and would prefer static and faster approaches like a diagram. For example, P2 reported that they are *"unsure whether the visual history is a better memory assistant than just the static content"*. Two participants in the **In-person study group** acknowledged the value of being immersed in the analysis playback.

AP2. Support Action Recovery: Slightly more than half (8/17) of participants confirmed that undo and redo are crucial and necessary. In terms of the efficiency for action recovery, 4/17 participants preferred interactions for quick undo and redo, and 2/17 participants proposed the ability to undo or redo multiple steps at once.

AP3. Support Replication and Reproducibility: Several (5/17) participants provided feedback on branching investigation. Four of them found the Copy-It-Spatula feature is suitable for branching out the investigation. One of these participants mentioned that it would be useful if they can copy branches instead of a single node at the time. One participant expressed the importance of the continuity of the data exploration environment. Creating a new branch from scratch or copying the branch to a new environment, might break this continuity.

AP4. Support Annotation: A notable number of participants (6/17) confirmed that annotations is helpful and important for both individual and collaborative analysis. Several (4/17) participants endorsed the idea of speech-to-text annotations, as "*text input in VR is still not easy to use*" (P2) and "*it is convenient to use voice command instead of typing*" (P5).

AP5. Support Sharing and Collaboration: Five participants highlighted the usefulness of the replayed annotations. P12 stated, "*I like the shared annotations most as it not only saves the cognitive load of the previous analyst, but also share insights or hints to collaborators easily at the situated data locations*". Notably, all the five participants were in the **In-person study group** and no one in the **Video group** commented on the replayed annotations. Participants also would like a combination of overview (from the provenance timeline view) and the first person view (the interactive analysis view) during the playback.

EP1. Use embodied representation of data construct and interaction metaphor: Most participants (8 from the **In-person study group** and 1 from the **Video group**) reported that the embodied representations and interaction metaphors in the system are engaging and fun. For example, P15 indicated that "*assigning an action or object for a specific function is quite interesting*".

We found divergent viewpoints regarding the usability of current applied physical metaphors. 8/17 participants (7 from the **In-person study group** and 1 from the **Video group**) doubted the use of a lever metaphor for action recovery and suggested simpler and faster approaches such as buttons or gestures. However, 5/17 participants who were in the **In-person study group** appreciated the use of microphone metaphor reporting that it's natural for annotations.

EP2. Support seamless integration between temporal and three-dimensional spatial aspects: Most participants (9/17) commented that the multiple views that integrate temporal information into the three-dimensional environment are useful for the whole analysis process. Participants found that having both the interactive analysis view and the provenance timeline view is especially useful for supporting multiple levels of analysis. However, 3 participants from the **In-person study group** pointed out that the requirement of frequent teleportation between the views could be confusing and exhausting.

EP3. Support environmental features to aid spatial memory: Six participants reported that different environments are useful for recalling previous analysis. For example, P16 confirmed that "*different contexts cue different memories*". Participants had different opinions on sounds and landmarks. For instance, 4/17 participants acknowledged the usefulness of sound in recalling memories and supporting the sense of presence, whereas some others (4/17) said that the sound could be irritating. A couple of participants (2/17) found landmarks are helpful to recall memories but another 2/17 reported that they did not make any difference. Most participants who had strong opinions on the sounds and landmarks are from the **In-person study group**.

Some (2/17) participants reported that they didn't notice the sounds and landmarks. Others (3/17) mentioned that they did not have enough experience to provide opinions on different environments, sounds, and landmarks. In the Video group, this is due to the fact that they did not try the prototype first hand. However, in the In-person study group, it could be due to the time limitation during the study.

7 DISCUSSION

Our work is, to our knowledge, the first explicit attempt to explore embodied provenance for immersive sensemaking. Results of the qualitative evaluation show the potential of our framework and system. Participants were receptive to the embodied representations and interactions we presented, and see many of our proposed features as engaging and useful ways to perform data analysis and analytic provenance. In this section, we highlight implications, limitations, and possible analytics scenarios in other immersive analytics systems.

7.1 Implications for proposed embodied provenance framework

Physical tool metaphors are engaging but should be simple and the alignment with the user's personal experience is beneficial. We designed a set of physical tools, including the lever, the helm, the spatula and the microphone, to support **EP1**. The idea of using physical tool metaphors is recognised by previous work that users can have a sense of engagement by directly manipulating the physical objects [5, 10]. The engagement advantage of our approach is also confirmed by participants' feedback. However, we did receive some criticisms over the physical tools used in our prototype system. An apparent example is the lever for action recovery. Most participants found it confusing and complex because they are familiar with a simple click to undo or redo actions in a desktop-based application and a lever does not conform to their perceived physical metaphor for action recovery. A positive example in our prototype for an appropriate use of physical metaphors is the microphone for annotations. This metaphor received positive reviews because it is natural and common to use a microphone to record audio in the real world. We learnt from the feedback that physical tool metaphors are engaging and could be useful to support embodied provenance, but the interactions should be simple and resemble familiar tools otherwise they may hinder functionality. Future research can benefit from exploring alternative physical metaphors that are common and simple to use for interactions such as undo and redo.

Integrating temporal information into large three-dimensional space with multiple views is useful and supporting smooth navigation is crucial. We provided multiple views to integrate temporal and 3D spatial aspects in the prototype system. These views are reported to be useful in supporting **EP2** and the whole analysis. However, some participants suggested better transitions between views as the current prototype requires frequent change of viewpoint during the navigation. Such statements of lacking smooth transition might confirm and potentially offer validation of the importance of EP2 regarding the seamless integration. Future research will benefit from the exploration of the navigation approach for embodied provenance systems. An example would be adding a see-through portal in the provenance timeline view so that users can to observe the detailed tabletops and the overview at the same time.

Themed environments with certain landmarks could potentially support spatial memory but non-visual elements should be used carefully. We offered six themed environments with landmarks and sounds to support **EP3**. Confirming prior work by Luo et al. [30], some participants found different environments with landmarks and sounds are useful for recalling analysis. However, some indicated that sounds could be annoying when they focused on the analysis. We learnt from these comments that although sounds could be helpful, the proper audio design is important to ensure that the ambient is not distracting to the user.

The mixed feedback of the environment could be attributed to the study design. In our study, participants spent most of their time within one environment and had limited exposure to different types of provided environments. Future research will benefit from an investigation of spatial memory aspect. In particular, it would be interesting to see whether there are ways to better support memory recall through the manipulation of the environmental factors such as visual elements or sounds.

7.2 Limitations

While the contribution of our work resides within the proposed embodied provenance concept and framework, our work requires the scaffolding of an immersive analytics system to provide the necessary functionality to demonstrate the concept. We chose to use the TimeTables [50] system for this purpose. While the results show promise of our prototype system in supporting embodied provenance, it is restricted to one specific application (based on TimeTables), for analysis of spatial-temporal data represented by tabletops and blocks using space-time cubes. This is a limitation that could be remedied by integrating our embodied provenance framework in other analytical systems and generalising the features we explored to other tasks and data representations in the future.

7.3 Possible Analytic Scenarios

To see how our proposed embodied provenance concepts can be applied to systems beyond TimeTables, we envision two analysis scenarios varying in the type of data: 1) abstract multidimensional and 2) network datasets.

Scenario 1: DataHop system with a multidimensional dataset of movies. As shown in Figure 2, DataHop partially supports memory recovery and action recovery by spatially laying out analysis histories in the large 3D space and provides easy switching between different states.

The system could improve analytic provenance support by adding more memory recall **(AP1)**, undo, and redo **(AP2)** options. DataHop can also add the ability to support annotation **(AP4)** and sharing **(AP5)**. Users should be able to store any annotations during their exploration and share their whole analysis with others by saving all states in the environment. As massive analysis could be conducted in the unlimited 3D space, landmarks, and other environmental features could be added in DataHop to aid spatial memory **(EP3)**.

The features of our prototype system can easily be adapted to the DataHop system. DataHop already has a main view for data analysis and a mini-map view for navigation, a provenance timeline view can be added to show the analysis history. Instead of tables in our current system, the analysis states will be presented as data panels created in DataHop. Users can interact with the data panels on the timeline to perform the Replay-It feature in recalling previous analysis and the Copy-It-Spatula feature in replicating and reproducing the analysis process. The actions in DataHop for analysis mainly include filtering data panels and grouping, distributing, and filtering the data within data panels. Users could recover the last performed action using the Do-It-Lever feature and add notes for any insights and findings using our Note-Down-Mic feature in our system.

Scenario 2: Multivariate network dataset. Three-dimensional network visualisation can be spatially presented in a large virtual space and the nodes can be directly manipulated **(EP1 and EP2)**. Data analysis actions include zoom, rotate, and select, among many others. The selection of different clusters forms different analytic states. By following the conceptual framework, users could be able to navigate to different states, recover the last actions **(AP2)** and recall previous analysis process **(AP1)**. They could also add annotations **(AP4)** and share analyses with others **(AP5)**.

In terms of the application of our prototype features to analyse network data, each cluster can be treated as the tabletop in our case. Once a cluster is explored, a thumbnail cluster could be added to the provenance timeline view. Users could use Replay-It and Copy-It-Spatula to recall and replicate analytic states. Users can also undo or redo the actions like rotating the graph using our Do-It-Lever, and use the Note-Down-Mic to record and save any insights.

7.4 Future Work

In future work, we aim to improve the implementation of the prototype system by adding new features and interactions based on participants' feedback. For instance, we could present the buttons for operating the playback, e.g. the pause button, with users so that they can be manipulated anywhere at any time. For a faster recall of previous analysis, we would like to add options to switch the speed of the playback and rewind the playback, and would also like to add action diagrams for tabletops in the timeline. We would work on enabling users to copy a branch of the tabletop tree rather than selecting tabletops one by one, having options to include child tabletops when copying the parent tabletop. We could also provide both text and audio output for annotations and allow users to undo and redo annotations as well.

One interesting challenge for future work will be to study how to integrate abstract provenance representations along with the spatial table arrangements to provide a more concise and efficient view for understanding detailed historical information. We also wish to explore the possibilities for synchronous and multi-user collaboration.

Evaluating these additional features and interactions would allow us to further reflect and refine upon our proposed conceptual framework for embodied provenance.

8 CONCLUSION

In this paper, we proposed a concept of embodied provenance, the use of three-dimensional space and embodied interactions in supporting recalling, reproducing, annotating and sharing analysis history in immersive environments. We defined the conceptual framework of embodied provenance by highlighting design criteria on analytic provenance from prior work and identifying embodied provenance properties. We presented a prototype system to demonstrate the concept with multiple data views, spatial environments and novel embodied interactions. The idea of embodied provenance and several features of our prototype system show promise according to the results of a qualitative evaluation with 17 participants with expertise in related areas. We hope our contributions assist future design of immersive analytic tools for embodied provenance that allow analytic provenance to play a stronger role in the sensemaking process.

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